Yiwen Hu

Assistant Professor
Department of Computer Science and Electrical Engineering
University of Maryland, Baltimore County

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Research Interests

My research interests span wireless networking, mobile systems (5G and beyond), cybersecurity, Internet of Things (IoT), and vehicle-to-everything (V2X). My recent work focuses on enhancing the reliability and security of critical network infrastructure from design to practice using formal methods, AI, and empirical experimentation.

Education

Michigan State University (MSU), East Lansing, MI — August 2018 – August 2025
Ph.D, Computer Science
Advisor: Prof. Guan-Hua Tu, & Prof. Li Xiao

Zhejiang University (ZJU), Hangzhou, China — September 2014 – June 2018
B.E., Digital Media Technology
College of Computer Science & Technology

Work Experience

University of Maryland, Baltimore County, Baltimore, MD — August 2025 – Present
Assistant Professor
Department of Computer Science and Electrical Engineering

Michigan State University (MSU), East Lansing, MI — August 2018 – May 2025
Research Assistant
Lab Director: Prof. Guan-Hua Tu, & Prof. Li Xiao

Selected Publications

[MobiCom'24] **Yiwen Hu**, Min-Yue Chen, Haitian Yan, Chuan-Yi Cheng, Guan-Hua Tu, Chi-Yu Li, Tian Xie, Chunyi Peng, Li Xiao, and Jiliang Tang. "Uncovering Problematic Designs Hindering Ubiquitous Cellular Emergency Services Access." In Proceedings of the 30th Annual International Conference On Mobile Computing And Networking (MobiCom). 2024. Washington, D.C., USA. (Acceptance rate: 55/288 = 19.09%, Winter run) [Media Coverage: MSU Today]

[MobiCom'24] Jingwen Shi, Sihan Wang, Min-Yue Chen, Guan-Hua Tu, Tian Xie, Man-Hsin Chen, **Yiwen Hu**, Chi-Yu Li, and Chunyi Peng. "IMS is Not That Secure on Your 5G/4G Phones." In Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (MobiCom), pp. 513-527. 2024. Washington, D.C., USA. (Acceptance rate: 48/207 = 23.19%, Summer run)

[ToN'24] Min-Yue Chen*, **Yiwen Hu*(*Co-primary)**, Guan-Hua Tu, Chi-Yu Li, Sihan Wang, Jingwen Shi, Tian Xie et al. "Taming the Insecurity of Cellular Emergency Services (9-1-1): From Vulnerabilities to Secure Designs." IEEE/ACM Transactions on Networking (ToN). 2024. **[Impact Factor: 3.79]**

[GetMobile'23] **Yiwen Hu**, Min-Yue Chen, Guan-Hua Tu, Chi-Yu Li, Sihan Wang, Jingwen Shi, Tian Xie et al. "Unveiling the Insecurity of Operational Cellular Emergency Services (911): Vulnerabilities, Attacks, and Countermeasures." ACM GetMobile Magazine: Mobile Computing and Communications (GetMobile), Volume:27, Issue: 1, pp. 39-43, March, 2023. Invited Article & Research Highlights.

[ToN'23] Sihan Wang, Tian Xie, Min-Yue Chen, Guan-Hua Tu, Chi-Yu Li, Xinyu Lei, Po-Yi Chou, Fucheng Hsieh, **Yiwen Hu**, Li Xiao, and Chunyi Peng. "Dissecting Operational Cellular IoT Service Security: Attacks and Defenses." IEEE/ACM Transactions on Networking (ToN), 2023. **[Impact Factor: 3.79]**

[MobiCom'22] **Yiwen Hu**, Min-Yue Chen, Guan-Hua Tu, Chi-Yu Li, Sihan Wang, Jingwen Shi, Tian Xie et al. "Uncovering insecure designs of cellular emergency services (911)." In Proceedings of the 28th Annual International Conference on Mobile Computing And Networking (MobiCom), pp. 703-715. 2022. Sydney, Australia. (Acceptance rate: 56/314 = 17.83%) Best Community Paper Award - Runner Up, AT&T Security Award, ACM SigMobile Research Highlights [Media Coverage: MSU Today, Wilx TV, Wood TV, WILS Morning Wake-Up, Homeland Security News, ISSSource, EurekAlert]

[MobiCom'21] Sihan Wang, Guan-Hua Tu, Xinyu Lei, Tian Xie, Chi-Yu Li, Po-Yi Chou, Fucheng Hsieh, **Yiwen Hu**, Li Xiao, and Chunyi Peng. "Insecurity of operational cellular IoT service: new vulnerabilities, attacks, and countermeasures." In Proceedings of the 27th Annual International Conference on Mobile Computing and Networking (MobiCom), pp. 437-450. 2021. New Orleans, United States.(Acceptance rate: 59/297 = 19.9%)

[CodaSpy'21] **Yiwen Hu**, Sihan Wang, Guan-Hua Tu, Li Xiao, Tian Xie, Xinyu Lei, and Chi-Yu Li. "Security threats from bitcoin wallet smartphone applications: Vulnerabilities, attacks, and countermeasures." In Proceedings of the Eleventh ACM Conference on Data and Application Security and Privacy (CodaSpy), pp. 89-100. 2021. Taking place virtually. (Acceptance rate: 26/106=24.53%)[Media Coverage: StudyFinds, MSU Today, EurekAlert, HelpNetSecurity]

Full Publications (➤ Link: Google Scholar Profile)

- **Conference Papers**

[MobiCom'24] **Yiwen Hu**, Min-Yue Chen, Haitian Yan, Chuan-Yi Cheng, Guan-Hua Tu, Chi-Yu Li, Tian Xie, Chunyi Peng, Li Xiao, and Jiliang Tang. "Uncovering Problematic Designs Hindering Ubiquitous Cellular Emergency Services Access." In Proceedings of the 30th Annual International Conference On Mobile Computing And Networking (MobiCom). 2024. Washington, D.C., USA. (Acceptance rate: 55/288 = 19.09%, Winter run) [Media Coverage: MSU Today]

[MobiCom'24] Jingwen Shi, Sihan Wang, Min-Yue Chen, Guan-Hua Tu, Tian Xie, Man-Hsin Chen, **Yiwen Hu**, Chi-Yu Li, and Chunyi Peng. "IMS is Not That Secure on Your 5G/4G Phones." In Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (MobiCom), pp. 513-527. 2024. Washington, D.C., USA. (Acceptance rate: 48/207 = 23.19%, Summer run)

[CNS'23] Jingwen Shi, Tian Xie, Guan-Hua Tu, Chunyi Peng, Chi-Yu Li, Andrew Hou, Sihan Wang, **Yiwen Hu**, Xinyu Lei, Min-Yue Chen, Li Xiao, and Xiaoming Liu "When Good Turns Evil: Encrypted 5G/4G Voice Calls Can Leak Your Identities." In 2023 IEEE Conference on Communications and Network Security (CNS), pp. 1-9. IEEE, 2023. Olando, Florida

[IJCNN'23] Xinyang Li, Xinyu Lei, Yantao Li, Huafeng Qin, and **Yiwen Hu**. "Towards Inference of Original Graph Data Information from Graph Embeddings." In 2023 International Joint Conference on Neural Networks (IJCNN), pp. 1-10. IEEE, 2023.

[MobiCom'22] **Yiwen Hu**, Min-Yue Chen, Guan-Hua Tu, Chi-Yu Li, Sihan Wang, Jingwen Shi, Tian Xie et al. "Uncovering insecure designs of cellular emergency services (911)." In Proceedings of the 28th Annual International Conference on Mobile Computing And Networking (MobiCom), pp. 703-715. 2022. Sydney, Australia. (Acceptance rate: 56/314 = 17.83%) Best Community Paper Award - Runner Up, AT&T Security Award, ACM SigMobile Research Highlights [Media Coverage: MSU Today, Wilx TV, Wood TV, WILS Morning Wake-Up, Homeland Security News, ISSSource, EurekAlert]

[MobiCom'21] Sihan Wang, Guan-Hua Tu, Xinyu Lei, Tian Xie, Chi-Yu Li, Po-Yi Chou, Fucheng Hsieh, **Yiwen Hu**, Li Xiao, and Chunyi Peng. "Insecurity of operational cellular IoT service: new vulnerabilities, attacks, and countermeasures." In Proceedings of the 27th Annual International Conference on Mobile Computing and Networking (MobiCom), pp. 437-450. 2021. New Orleans, United States.(Acceptance rate: 59/297 = 19.9%)

[CodaSpy'21] **Yiwen Hu**, Sihan Wang, Guan-Hua Tu, Li Xiao, Tian Xie, Xinyu Lei, and Chi-Yu Li. "Security threats from bitcoin wallet smartphone applications: Vulnerabilities, attacks, and countermeasures." In Proceedings of the Eleventh ACM Conference on Data and Application Security and Privacy (CodaSpy), pp. 89-100. 2021. Taking place virtually. (Acceptance rate: 26/106=24.53%)[Media Coverage: StudyFinds, MSU Today, EurekAlert, HelpNetSecurity]

- **Journal/Magazine Articles**

[GetMobile'25] Jingwen Shi, Sihan Wang, Min-Yue Chen, Guan-Hua Tu, Tian Xie, Man-Hsin Chen, **Yiwen Hu**, Chi-Yu Li, and Chunyi Peng. "IMS is Not That Secure on Your 5G/4G Phones." ACM GetMobile Magazine: Mobile Computing and Communications, Volume 29, Issue 1, pp. 20-24, April 2025. Invited Article & Research Highlights.

[TOSN'24] Yantao Li, Xinyang Li, Xinyu Lei, Huafeng Qin, **Yiwen Hu**, and Gang Zhou. "On the Inference of Original Graph Information from Graph Embeddings." ACM Transactions on Sensor Networks (TOSN). 2024. **[Impact Factor: 3.9]**

[ToN'24] Min-Yue Chen*, **Yiwen Hu*(*Co-primary)**, Guan-Hua Tu, Chi-Yu Li, Sihan Wang, Jingwen Shi, Tian Xie et al. "Taming the Insecurity of Cellular Emergency Services (9-1-1): From Vulnerabilities to Secure Designs." IEEE/ACM Transactions on Networking (ToN). 2024. **[Impact Factor: 3.79]**

[GetMobile'23] **Yiwen Hu**, Min-Yue Chen, Guan-Hua Tu, Chi-Yu Li, Sihan Wang, Jingwen Shi, Tian Xie et al. "Unveiling the Insecurity of Operational Cellular Emergency Services (911): Vulnerabilities, Attacks, and Countermeasures." ACM GetMobile Magazine: Mobile Computing and Communications (GetMobile), Volume:27, Issue: 1, pp. 39-43, March, 2023. Invited Article & Research Highlights.

[ToN'23] Sihan Wang, Tian Xie, Min-Yue Chen, Guan-Hua Tu, Chi-Yu Li, Xinyu Lei, Po-Yi Chou, Fucheng Hsieh, **Yiwen Hu**, Li Xiao, and Chunyi Peng. "Dissecting Operational Cellular IoT Service Security: Attacks and Defenses." IEEE/ACM Transactions on Networking (ToN), 2023. **[Impact Factor: 3.79]**

[CSUR'23] Xiao Zhang, Griffin Klevering, Xinyu Lei, **Yiwen Hu**, Li Xiao, and Guan-Hua Tu. "The security in optical wireless communication: A survey." ACM Computing Surveys (CSUR) 55, no. 14s. 2023: 1-36. **[Impact Factor: 14.324]**

Honors and Awards

• MSU CSE Recognition for Outstanding Research	2025
• ACM GetMobile'25 Research Highlights	2025
• MSU Dissertation Completion Fellowship	2024
• **ACM SigMobile'24 Research Highlights** (First-Author Paper)	2024
• MSU Graduate School Travel Grant	2024
• **ACM GetMobile'23 Research Highlights** (First-Author Paper)	2023
• **911 CVD report contributed to GSMA standard changes** (First-Author Paper)	2023
• **AT&T Security Award** (First-Author Paper)	2023
• MSU Graduate Fellowship	2023
• **ACM MobiCom'22 Best Community Paper Award Runner-Up** (First-Author Paper)	2022
• ACM MobiCom'21 Student Travel Grant	2022
• Carl V. Page Graduate Memorial Fellowship	2022

Proposal Preparation Experience

- **Collaborative Research: SaTC: Core: Medium: Safeguarding Next-Generation Emergency Services (NG-9-1-1) over Cellular Networks: From Design to Practice**
Sponsor: **National Science Foundation**
Investigators: Guan-Hua Tu (Lead PI) (47.13%), Jiliang Tang, Li Xiao, and Chunyi Peng (PI at Purdue University).
Amount: $1,200,000
Period: 07/01/2023 - 06/30/2026

- **NeTS: Small: Exploring the Non-Standardized Polices, Operations, and Requirements for 5G Cellular Networks and Beyond: Advancing the Modeling, Tools, and Evaluation**
Sponsor: **National Science Foundation**
Investigators: Guan-Hua Tu (PI) (65%), and Hui Liu.
Amount: $599,995
Period: 10/01/2023 - 09/30/2026

Teaching Experience

CMSC 841: Computer Networks, Instructor — Fall 2025
Undergraduate Course, CSEE, UMBC, (Teaching a 40-student course, including lectures, assignments, exams, and office hours.)

CSE 824: Advanced Computer Networks and Communications, by Prof. Li Xiao — Fall 2024
Graduate Course, Department of Computer Science and Engineering, MSU
Guest Lecturer (Providing guest lectures for a 50-student course)

CSE 480 Database Systems, by Prof. Manni Liu & Prof. Hamid Karimian — Summer 2024
Undergraduate Course, Department of Computer Science and Engineering, MSU
Teaching Assistant (Office hour, grading, online Q&A for of a 100-student course)

CSE 425 Introduction to Computer Security, by Prof. Guan-Hua Tu — Spring 2022
Undergraduate Course, Department of Computer Science and Engineering, MSU
Teaching Assistant (Office hour, grading, online Q&A for of a 100-student course)

CSE 325 Computer System, by Prof. Philip McKinley — Fall 2021
Undergraduate Course, Department of Computer Science and Engineering, MSU
Teaching Assistant (Office hour, grading, online Q&A for of a 200-student course)

CSE 325 Computer System, by Prof. Mark McCullen — Summer 2021
Undergraduate Course, Department of Computer Science and Engineering, MSU
Teaching Assistant (Office hour, grading, online Q&A for of a 200-student course)

CSE 325 Computer System, by Prof. Mark McCullen — Spring 2021
Undergraduate Course, Department of Computer Science and Engineering, MSU
Teaching Assistant (Office hour, grading, online Q&A for of a 200-student course)

CSE 325 Computer System, by Prof. Mark McCullen — Fall 2020
Undergraduate Course, Department of Computer Science and Engineering, MSU
Teaching Assistant (Office hour, grading, online Q&A for of a 200-student course)

CSE 325 Computer System, by Prof. Robert James — Summer 2020
Undergraduate Course, Department of Computer Science and Engineering, MSU
Teaching Assistant (Office hour, grading, online Q&A for of a 200-student course)

CSE 320 Object-Oriented Software Development, Prof. Charles Owen — Spring 2020
Undergraduate Course, Department of Computer Science and Engineering, MSU
Teaching Assistant (Office hour, grading, online Q&A for of a 200-student course)

Media Coverage

- MSUToday(2025). “MSU researchers identify why 911 calls are delayed, failed, or dropped”. March 27.
- ISSSource (2023). “Boosting Security For Cellular Emergency Calls”. May 1.
- TheWILS Morning Wake-Up (2023). “The National Science Foundation awarding MSU just over 1 million dollars, what are they using the money on?”. April 28.
- Homeland Security News Wire (2023). “Making Emergency Calls More Secure”. April 27.
- WoodTV(2023). “MSU researchers get $1.2M to improve 911 call security”. April 27.
- EurekAlert (2023). “MSU researchers to lead security improvements for cellular 911 calls with $1.2M NSF grant”. April 27.
- Wilx TV(2023). “MSU researchers awarded $1.2 million grant to fix 911 cyber security issues”. April 25.
- MSUToday(2023). “Making emergency calls more secure”. April 24.
- HelpNetSecurity (2021). “Bitcoin Security Rectifier app aims to make Bitcoin more secure”. May 10.
- StudyFinds (2021). “Bitcoin wallet apps are too vulnerable to use safely, researchers warn”. May 5.
- MSUToday(2021). “Making Bitcoin more secure”. May 4.
- EurekAlert (2021). “New app makes Bitcoin more secure”. May 4.

Invited Talks and Presentations

- *Powering the Infrastructure for Critical Services: Ensuring Secure and Reliable Emergency Communications over Cellular Networks* — 2025
The University of Minnesota Twin Cities, Minneapolis, MN

- *Powering the Infrastructure for Critical Services: Ensuring Secure and Reliable Emergency Communications over Cellular Networks* — 2025
University of Iowa, Iowa City, IA

- *Powering the Infrastructure for Critical Services: Ensuring Secure and Reliable Emergency Communications over Cellular Networks* — 2025
The University of Maryland, Baltimore County, Baltimore, MD

- *Powering the Infrastructure for Critical Services: Ensuring Secure and Reliable Emergency Communications over Cellular Networks* — 2025
The University of Nevada Reno, Reno, NV

- *Powering the Infrastructure for Critical Services: Ensuring Secure and Reliable Emergency Communications over Cellular Networks* — 2025
Wayne State University, Detroit, MI

- *Uncovering Problematic Designs Hindering Ubiquitous Cellular Emergency Services Access* — 2024
MobiCom 2024 Paper Presentation, Washington, D.C.

- *5G/4G Mobile Networks Architectures and Services* — 2024
MSU Graduate Course CSE 824, Michigan

- *Advanced Research on Mobile Networks and Services* — 2024
MSU Graduate Course CSE 824, Michigan

- *Security Threats from Bitcoin Wallet Smartphone Applications* — 2022
MSU Graduate Seminar, Michigan

- *Insecurity of Operational Cellular IoT Services* — 2022
MobiCom 2021 Paper Presentation, New Orleans

Professional Services

Conference Publicity Chair

- IEEE Conference on Dependable and Secure Computing (DSC) 2025

Conference Technical Program Committee

- IEEE Conference on Dependable and Secure Computing (DSC) 2025
- IEEE Conference on Communications and Network Security (CNS) 2025
- IEEE International Conference on Parallel and Distributed Systems (ICPADS) 2025
- IEEE The International Joint Conference on Neural Networks (IJCNN) 2024

Journal Reviewer

- IEEE Transactions on Dependable and Secure Computing (TDSC) 2025
- ACM Journal on Autonomous Transportation Systems (ACM JATS) 2025
- IEEE Transactions on Reliability (IEEE TR) 2025
- ACM Transactions on Internet of Things (ACM TIOT) 2025
- IEEE Transactions on Networking (IEEE TNET) 2025
- IEEE Transactions on Networking (IEEE TNET) 2024
- IEEE Transactions on Reliability (IEEE TR) 2024
- Neural Computing and Applications (NCAA) 2024
- IEEE Transactions on Networking (IEEE TNET) 2023

Conference Reviewer

- IEEE International Conference on Computer Communications (INFOCOM) 2026
- IEEE Conference on Communications and Network Security (CNS) 2025
- IEEE Conference on Dependable and Secure Computing (DSC) 2025
- IEEE International Conference on Parallel and Distributed Systems (ICPADS) 2025
- IEEE The International Joint Conference on Neural Networks (IJCNN) 2024

External Reviewer

- IEEE International Conference on Computer Communications (INFOCOM) 2024
- ACM 5G and Beyond Network Measurements, Modeling, and Use Cases (5G-MeMU) 2024
- IEEE Military Communications Conference (MILCOM) 2024
- IEEE International Conference on Computer Communications (INFOCOM) 2023
- International Conference on Neural Information Processing (ICONIP) 2023
- IEEE International Conference on Network Protocols (ICNP) 2023
- IEEE Conference on Communications and Network Security (CNS) 2023
- IEEE International Conference on Computer Communications and Networks (ICCCN) 2023
- IFIP Networking Conference (IFIP) 2023
- IEEE International Conference on Communications (ICC) 2023
- IEEE Conference on Big Data (BigData) 2022
- IEEE International Conference on Computer Communications (INFOCOM) 2021
- IEEE Vehicular Networking Conference (VNC) 2020
- IEEE International Conference on Computer Communications (INFOCOM) 2020
- IEEE Global Communications Conference (GLOBECOM) 2019
- IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS) 2019

Members and Services

- **Student member**, Association for Computing Machinery (ACM) 2024
- **Student member**, Association for Computing Machinery (ACM) 2023
- **Volunteer**, The 27th Annual International Conference on Mobile Computing and Networking 2021